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PROJECT REPORT

SMART MEDICINE BOX WITH ROBOT ASSISTANCE

Submitted in partial fulfilment of the requirements for the degree of Bachelor of Engineering

Computer Science and Engineering (Data Science)

Submitted by

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ST. JOHN COLLEGE OF ENGINEERING & MANAGEMENT
UNIVERSITY OF MUMBAI
2026–2027

CERTIFICATE OF APPROVAL

This is to certify that Om Kadam (EU1258047), Rishabh Sonawane (EU1258001), Pravin Tiwari (EU1258039), and Om Tripathi (EU1258052) are bona fide students of the Bachelor of Engineering programme in Computer Science and Engineering (Data Science), St. John College of Engineering and Management, Mumbai. They have prepared the project report entitled **"Smart Medicine Box with Robot Assistance"** under the guidance of Prof. Sachin Sase.

Project Guide	Head of Department
Signature: _____	Signature: _____
Name: Prof. Sachin Sase	Name: Mrs. Joslyn Gracias
Designation: Assistant Professor	Designation: Head of Department

Principal: Dr. Kamal Shah

Signature: _____

Date: _____

DECLARATION

We declare that this written submission represents our ideas in our own words. Wherever the ideas or words of others have been included, we have cited or referenced the original sources. We further declare that the report describes the current prototype honestly and does not present planned future features as completed work.

Student Name	Signature	Date
Om Kadam	_____	_____
Rishabh Sonawane	_____	_____
Pravin Tiwari	_____	_____
Om Tripathi	_____	_____

Place: _____

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Group Members

- Om Kadam
- Rishabh Sonawane
- Pravin Tiwari
- Om Tripathi

ABSTRACT

The Smart Medicine Box with Robot Assistance is a hardware prototype developed to make scheduled medicine reminders more noticeable through simple audible and visual alerts. The project focuses on helping users notice a reminder at the configured time. In the current demonstration, when the configured time for Compartment 1 is reached, a buzzer sounds and an LED blinks. These two indicators provide a clear reminder without claiming to verify whether the medicine has actually been taken.

The prototype combines the medicine-reminder unit with a robot assembly. The system uses an ESP32 controller, a buzzer, an LED indicator, an L298N motor driver, two 12 V, 300 RPM DC motors, a mini servo, an acrylic body, and 3D-printed wheels and face components. Together, these components demonstrate the basic reminder function and the physical robot platform. The project brings together embedded control, electronic output indicators, and a small robotic structure in one prototype.

Keywords: Smart medicine box, medicine reminder, ESP32, buzzer, LED indicator, robot assistance, L298N, assistive technology.

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ABBREVIATIONS AND SYMBOLS

Term	Meaning
ESP32	Microcontroller development platform
LED	Light-Emitting Diode
DC	Direct Current
USB	Universal Serial Bus
RPM	Revolutions Per Minute
L298N	Dual H-bridge motor-driver module
BOM	Bill of Materials
SDG	Sustainable Development Goal
V	Volt

CHAPTER 1: INTRODUCTION

1.1 Background

Taking medicine at the prescribed time is important, but busy routines and distractions can cause missed doses. A buzzer and blinking LED make a scheduled reminder more noticeable. This project combines that reminder concept with a small robot prototype. It is an educational prototype, not a medical device or a replacement for professional instructions.

1.2 Problem Statement

Ordinary medicine boxes mainly store medicines and may not actively alert users at a scheduled time. The project addresses this issue through a buzzer and LED reminder while exploring a robot-assisted concept for future development.

1.3 Need of the Project

- Provide an audible and visual alert at a scheduled time.
- Organise medicines in designated compartments.
- Demonstrate an affordable embedded-system concept.
- Build a foundation for future robot movement and communication.

1.4 Objectives

- Assemble a medicine box with compartments.
- Trigger a buzzer and blinking LED for Compartment 1.
- Build a small robot using available controller and motor hardware.
- Document current functions and plan safe future integration.

1.5 Scope and Expected Outcomes

Current scope includes the Compartment 1 alert and a basic wired robot prototype. Autonomous dispatch, mobile notifications, dose logging and multi-compartment scheduling are outside the verified scope. Expected outcomes are a demonstrable reminder, a reusable robot chassis and a documented plan for testing and improvement.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

Medicine reminders commonly use alarms, indicator lights or mobile notifications. Mobile robots use a controller, motor driver, motors and a power source. This project combines these ideas at prototype level. The table below is a technology-oriented comparison, not a formal survey of research papers.

2.2 Related-System Comparison

Approach	Strengths	Limitations
Basic pill organiser	Low-cost medicine separation	May not provide an active reminder
Alarm / phone reminder	Simple scheduled alert	Depends on device settings and user attention
Electronic reminder box	Can combine timer, buzzer and lights	Needs reliable power and correct setup
Mobile assistive robot	Can move a device or object	Needs motor control, navigation and safety checks
Proposed project	Combines reminder with robot concept	Current robot is wired/manual; automatic integration is pending

2.3 Research Gap

The practical gap explored is the integration of a scheduled reminder with a small, low-cost robot platform. The present prototype demonstrates the alert and basic wired robot elements; coordination between the medicine schedule and robot movement remains future work.

2.4 Novelty / Innovation

- Combines a medicine-compartment concept with a robot enclosure.
- Uses audible and visual signals for reminders.
- Defines a staged path from wired operation to possible wireless and autonomous features.

Research note: Add verified peer-reviewed papers if the department requires a formal literature survey. Do not cite papers that have not been read and checked.

CHAPTER 3: PROPOSED METHODOLOGY

3.1 Proposed System and Working Principle

The reminder checks the configured time for Compartment 1. When the due time occurs, the buzzer sounds and the LED blinks. The user follows their prescribed medicine instructions. The robot is demonstrated separately through its current wired USB/control arrangement; schedule-triggered autonomous travel is not implemented.

3.2 System Architecture

3.3 System Architecture / Block Diagram

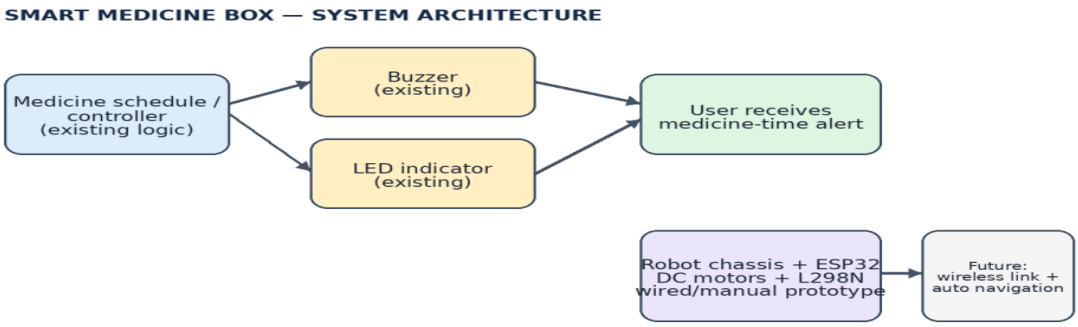


Figure 3.1: Conceptual architecture showing current reminder/robot elements and proposed integration.

3.3 Current Reminder Flow

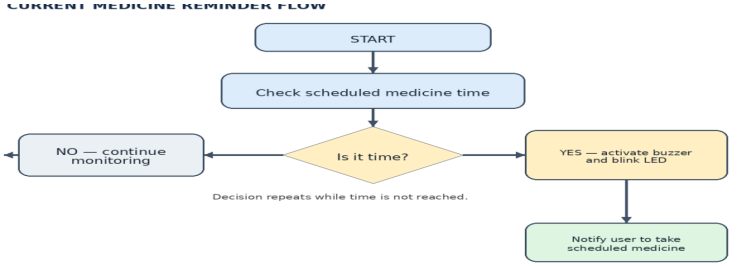


Figure 3.2: Current medicine reminder logic.

3.5 Proposed Future Robot Workflow



Figure 3.2: Current medicine reminder logic.

3.4 Methodology

- Define reminder and robot demonstration requirements.
- Assemble the acrylic enclosure and 3D-printed parts.
- Connect controller, motor driver, motors, servo and reminder components as applicable.
- Configure firmware, demonstrate each function separately and record actual test observations.

CHAPTER 3: COMPONENTS AND WORKFLOW

3.5 Components / Materials

Component	Specification / purpose	Quantity / status
ESP32	Robot control platform	1 reported; variant to confirm
L298N driver	Drives DC motors	Quantity to confirm
DC motors	12 V, 300 RPM	2
Mini servo	Neck/head movement	1
Acrylic sheets	Body / enclosure	As required
3D-printed parts	Wheels and face components	As required
Buzzer and LED	Audible and visual alerts	Confirm exact types/count
Medicine compartments	Separate medicine storage	Count to confirm
USB, wiring, power source	Connections and power	Confirm exact ratings

3.6 Main Component Functions

The ESP32 runs embedded control logic when programmed. The L298N interfaces controller signals with motor current. DC motors drive the wheels, while the mini servo can position the head/neck. The buzzer and LED provide the reminder outputs. Wireless capability of a controller does not by itself prove that wireless control is configured.

3.7 Safety Notes

- Never power 12 V motors directly from an ESP32 GPIO pin.
- Check motor-driver and power-supply ratings before operation.
- Disconnect power before changing wiring; secure exposed connections.
- Confirm the exact circuit, controller roles and firmware before final submission.

CHAPTER 4: PROJECT PLANNING

4.1 Work Breakdown Structure

Work package	Activities	Deliverable
Requirements	Define reminder and robot concept	Requirements list
Mechanical design	Plan enclosure and layout	Acrylic body and printed parts
Electronics	Connect controller, driver, motors and indicators	Assembled hardware
Firmware/control	Configure reminder and robot operation	Demonstration setup
Testing	Check alert and movement behaviour	Test log
Documentation	Prepare report, diagrams and photos	Project report

4.2 Proposed Semester Plan

Activity	Phase 1	Phase 2	Phase 3	Phase 4
Requirements / planning	●			
Mechanical assembly	●	●		
Electronics / controller setup		●	●	
Demonstration and testing			●	●
Documentation / presentation				●

This is a proposed plan, not a verified record of actual dates. Replace it with the team's real schedule if required.

4.3 Roles and Responsibilities

Team member	Suggested responsibility
Om Kadam	Coordination, requirements and integration
Rishabh Sonawane	Assembly support and interface documentation
Pravin Tiwari	Controller, motor-control and firmware support
Om Tripathi	Testing records, diagrams and report support

Allocation is a planning suggestion; edit it to reflect each member's actual contribution.

CHAPTER 5·7: FEASIBILITY, RESULTS AND CONCLUSION

5.1 Feasibility and Budget

The reminder and robot are technically feasible as a student prototype using common modules. Full automation would require reliable communication, motor control, power management, movement feedback and safety checks. The cost should be calculated from actual receipts or supplier quotations; no estimated prices are invented here.

Item	Quantity	Actual cost (INR)
ESP32 board	1 reported	Fill from receipt
L298N driver	Confirm	Fill from receipt
12 V DC motors	2	Fill from receipt
Servo, acrylic, printed parts	As required	Fill from receipt
Buzzer, LED, wiring, power	Confirm	Fill from receipt
Total	—	Calculate after entry

6.1 Testing Plan and Expected Results

Test	Expected behaviour	Observed result
Scheduled alert	Buzzer sounds and LED blinks at due time	Record during test
Before due time	No early alert under configured logic	Record during test
Robot wired operation	Responds to implemented wired control	Record during test
Schedule integration	No autonomous movement claimed	Not implemented

No measured results were supplied. Record observations after testing; do not claim accuracy or reliability percentages without repeated measurements.

6.2 Impact and Sustainability

The concept may support assistive-technology demonstrations and make medicine-time alerts more noticeable. It does not verify that medicine was taken. Reusable modules, repairable parts and responsible battery/e-waste disposal are encouraged. Potential SDG links: SDG 3 (Good Health and Well-Being), SDG 9 (Industry, Innovation and Infrastructure) and SDG 12 (Responsible Consumption and Production).

7. Conclusion and Future Scope

The prototype demonstrates a scheduled buzzer/LED reminder for Compartment 1 and a basic wired robot. Next steps are to verify the circuit, record test results, improve safe control, and investigate wireless control and schedule-triggered movement. Multiple schedules, mobile notifications and obstacle-aware navigation remain future improvements.

REFERENCES AND APPENDICES

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APPENDIX A: FINAL PROJECT PROTOTYPE PHOTOGRAPHS

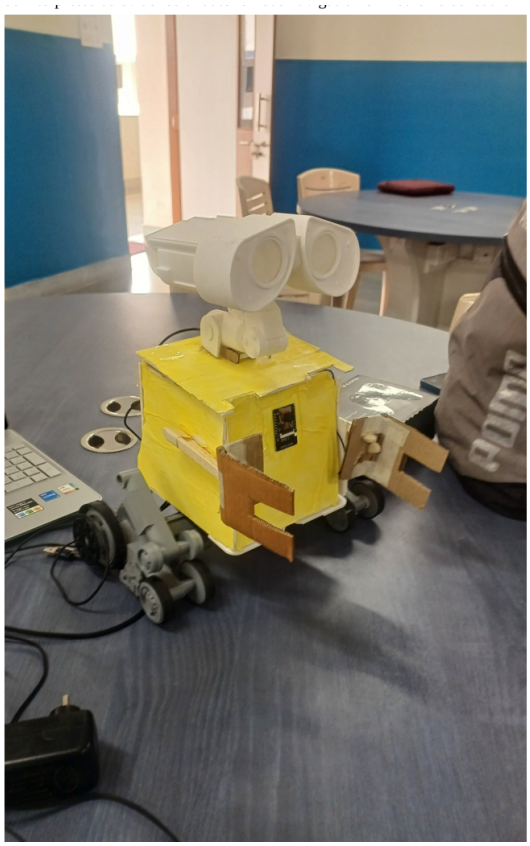


Figure A.1: Side view of the robot prototype.

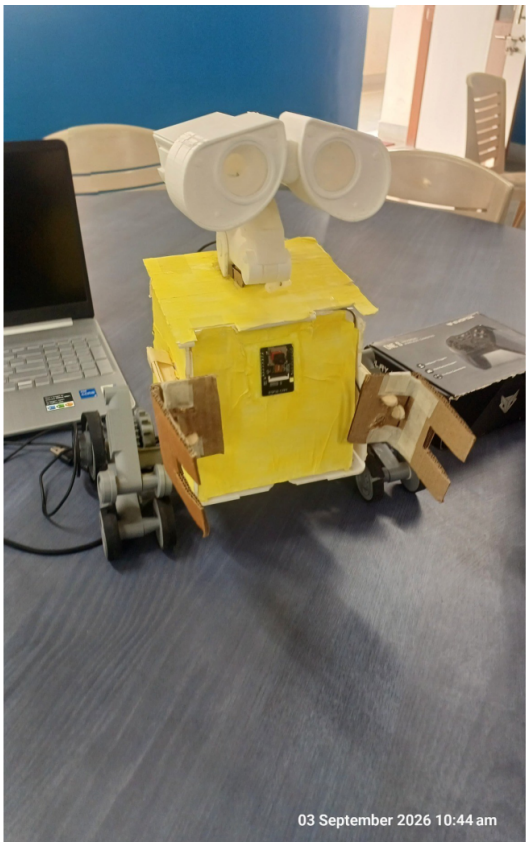


Figure A.2: Front view of the robot prototype.

Before final submission, confirm compartment count, controller and GPIO wiring, ESP32 variant, driver quantity, power ratings, actual test observations, component costs and required signatures.